Standard Operating Procedure (SOP) for Resolving Critical Hardware/Environmental Faults on Arista Access Nodes: MALL-ACC-21

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6152

Description

This SOP outlines the procedures for diagnosing and resolving critical hardware/environmental faults on Arista Access nodes, specifically MALL-ACC-21. The failure domain in question is related to ASIC memory parity errors, line card crashes, and power supply redundancy loss. These issues are typically rooted in the node's hardware design and can be exacerbated by environmental factors such as temperature fluctuations or power supply instability. The Arista EOS (Extensible Operating System) is designed to provide a robust and fault-tolerant platform for Access nodes, but in the event of a critical failure, it is essential to follow this SOP to ensure timely and effective resolution. The diagnostic and resolution commands outlined in this SOP are specific to the Arista CLI dialect and are intended to isolate the root cause of the failure and implement corrective actions.

Pre-requisite Checks

- Check: Verify that remote hands have been dispatched to the Malleswaram Node for physical assistance, if required.

- Check: Confirm that the redundant link status is active and functioning correctly.

- Check: Ensure that the node's environmental conditions, such as temperature and humidity, are within the recommended operating ranges.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware defects or wear and tear.	High
Line card crashes resulting from software or firmware issues, or environmental stressors.	Medium
Power supply redundancy loss due to hardware failure or configuration errors.	Low

Diagnostic Commands & Expected Outputs

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Command: show system internal error-monitor, Expected Output: Error-monitoring information, including error counts and timestamps., Description: This command retrieves error-monitoring information from the node, including error counts and timestamps. This can help identify patterns or recurring issues that may indicate the root cause of the failure.

Command: show hardware component, Expected Output: Hardware component information, including temperature and voltage readings., Description:
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This command provides information about the node's hardware components, including temperature and voltage readings. This can help identify potential environmental stressors or hardware defects.

Command: show inventory, Expected Output: Inventory information, including serial numbers and firmware versions., Description: This command retrieves inventory information from the node, including serial numbers and firmware versions. This can help identify any discrepancies or inconsistencies that may indicate a hardware or software issue.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card., Command: no switch 1/1-2, Description: This command removes the faulty line card from the node's configuration. Once removed, the line card can be replaced with a new one, and the node can be brought back online.

Step 2: Step: Reboot the node and verify that the power supply redundancy is functioning correctly., Command: reboot, Description: This command reboots the node and verifies that the power supply redundancy is functioning correctly. If the power supply redundancy is not functioning correctly, the node may not be able to boot up or may experience instability.

Step 3: Step: Update the node's firmware to the latest version., Command: no switch 1/1-2, Description: This command updates the node's firmware to the latest version. This can help resolve software or firmware issues that may be contributing to the failure.

Post-Check Verification

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Command: show system internal error-monitor, Expected Output: Error-monitoring information, including error counts and timestamps., Description: This command verifies that the node is no longer experiencing errors and that the error counts and timestamps are zero.

Command: show hardware component, Expected Output: Hardware component information, including temperature and voltage readings., Description: This command verifies that the node's hardware components are functioning correctly and that temperature and voltage readings are within the recommended ranges.

Command: show inventory, Expected Output: Inventory information, including serial numbers and firmware versions., Description: This command verifies that the node's inventory information is accurate and that serial numbers and firmware versions match the expected values.
```

Roll-back Procedures

- Step: Revert the node to its previous configuration., Command: configure replace switch 1/1-2, Description: This command reverts the node to its previous configuration. This can help restore the node to its previous state and prevent further issues.

- Step: Reboot the node and verify that it is functioning correctly., Command: reboot, Description: This command reboots the node and verifies that it is functioning correctly. If the node is not functioning correctly, the rollback procedure can be repeated until the node is stable.